

ARCTIC VILLAGE, AK, USA

WMO: 701945

Lat: 68.115N

Lon: 145.579W

Elev: 638

StdP: 93.90

Time Zone: -9.00 (NAK)

Period: 95-19

WBAN: 46405

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -43.9	(c) -42.1	(d) -35.1	(e) 0.1	(f) -32.4	(g) -34.1	(h) 0.2	(i) -29.8	(j) 8.8	(k) -14.6	(l) 7.4	(m) -15.8	(n) 0.4	(o) 40	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	22.6	12.9	21.2	12.4	19.0	11.3	14.0	20.3	13.1	19.1	12.3	17.7	3.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.4	9.0	15.2	10.6	8.6	15.2	9.0	7.7	14.9	41.1	20.0	38.7	18.9	36.6	17.6	16.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 7.9	(b) 6.8	(c) 5.6	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-7.5	-27.3	-22.8	-18.4	-10.1	3.4	12.4	13.9	9.5	3.1	-9.1	-21.5	-24.4
	DBStd	16.04	8.90	7.73	6.33	6.09	6.03	3.55	2.87	3.39	4.33	5.99	8.53	8.07
	HDD10.0	6657	1156	918	882	602	215	17	6	49	208	593	945	1067
	HDD18.3	9442	1414	1152	1140	852	464	179	139	273	457	851	1195	1325
	CDD10.0	258	0	0	0	0	9	89	126	34	1	0	0	0
	CDD18.3	2	0	0	0	0	0	1	1	0	0	0	0	0
	CDH23.3	29	0	0	0	0	0	11	16	3	0	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	2.0	1.2	1.2	1.7	2.2	2.6	3.0	2.8	2.7	2.3	1.7	1.2	1.4
Precipitation	PrecAvg	254	14	12	10	8	13	29	44	45	29	22	15	15
	PrecMax	348	38	33	21	29	28	62	73	75	58	38	43	35
	PrecMin	155	5	2	3	2	4	9	19	12	6	6	1	4
	PrecStd	43	8	8	5	6	5	14	13	16	14	8	10	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.9	-4.8	-1.0	7.3	20.8	24.8	25.3	22.9	15.1	5.3	-3.0	-4.7
		MCWB	-6.4	-5.9	-2.8	2.3	10.0	13.4	14.2	13.7	8.1	2.3	-4.4	-6.3
	2%	DB	-9.9	-7.5	-3.9	3.6	17.4	22.5	22.8	19.4	12.9	2.1	-5.9	-7.6
		MCWB	-10.0	-8.2	-5.9	-0.2	8.9	12.2	13.2	12.6	7.1	0.1	-6.8	-8.3
	5%	DB	-12.2	-9.9	-6.4	2.0	14.8	21.0	21.4	17.4	11.5	0.2	-7.6	-10.6
		MCWB	-12.7	-10.5	-8.0	-1.2	7.7	11.5	12.9	10.8	6.6	-1.1	-8.5	-11.0
	10%	DB	-14.0	-12.3	-8.7	-0.1	12.5	18.8	19.9	15.9	9.2	-1.3	-9.9	-13.0
		MCWB	-15.2	-12.9	-10.1	-2.5	6.5	10.5	12.2	10.1	5.6	-2.3	-10.3	-14.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.9	-4.7	-2.5	2.5	10.9	14.2	15.4	14.8	9.0	2.8	-4.2	-5.9
		MCDB	-5.3	-3.6	-1.5	7.1	18.3	21.6	23.5	21.3	13.5	4.5	-3.0	-4.3
	2%	WB	-8.7	-8.0	-5.6	0.1	9.6	13.2	14.4	13.0	7.9	0.4	-6.1	-8.0
		MCDB	-8.2	-7.3	-4.1	3.3	16.5	20.4	20.8	18.6	11.9	1.5	-5.2	-7.2
	5%	WB	-11.6	-10.2	-7.8	-1.2	8.2	12.3	13.6	11.7	7.0	-1.0	-8.2	-10.2
		MCDB	-11.6	-9.6	-6.3	1.4	14.1	18.8	19.4	16.0	10.6	0.0	-7.4	-9.6
	10%	WB	-13.3	-12.4	-9.8	-2.5	6.9	11.3	12.9	10.5	6.0	-2.4	-9.9	-13.1
		MCDB	-12.5	-12.1	-8.5	-0.1	12.0	17.6	18.4	14.6	9.3	-1.6	-9.2	-12.4
Mean Daily Temperature Range	5% DB	MDBR	8.2	8.9	12.5	12.2	9.5	9.1	8.7	8.4	7.7	7.0	7.8	8.9
		MCDBR	9.3	8.4	13.3	12.3	12.1	11.5	11.3	10.8	11.3	7.3	8.3	10.5
		MCWBR	8.9	7.9	11.8	9.2	6.4	4.8	4.8	5.4	6.5	5.8	8.0	10.2
	5% WB	MCDBR	9.4	8.3	12.8	11.3	11.3	9.9	9.9	9.3	9.7	6.6	8.2	10.5
		MCWBR	9.1	7.9	11.4	8.6	6.1	4.4	4.6	5.0	5.9	5.4	7.9	10.2
Clear-Sky Solar Irradiance	taub	0.160	0.219	0.239	0.307	0.319	0.340	0.356	0.320	0.288	0.223	0.155	N/A	
	taud	1.905	2.114	2.263	2.112	2.283	2.412	2.394	2.503	2.491	2.208	1.921	N/A	
	Ebn at Noon	134	604	826	832	871	861	830	831	770	592	121	0	
	Edh at Noon	8	44	71	112	107	98	97	77	59	39	7	0	
All-Sky Solar Radiation	RadAvg	0.04	0.56	2.35	4.68	5.80	5.91	4.97	3.62	2.11	0.77	0.11	0.00	
	RadStd	0.00	0.04	0.11	0.20	0.44	0.54	0.34	0.34	0.23	0.09	0.01	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page